Performance Improvement in Conventional 4-bit Static CMOS Carry Look-Ahead Adder by Modifying Carry-Generate and Propagate Terms

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***Abstract*—Conventional static CMOS design employing pull-up and pull-down network for implementing logic functions has been one of the most reliable and widely used design techniques over the years. However, high input impedance and transistor count remains a major problem which result in high silicon area, high power consumption and increased delay. This research focus on reducing power, delay and transistor count in Conventional Static CMOS based 4-bit Carry Look Ahead adder. The conventional static CMOS XOR and gates are replaced by Gate Diffusion Input technique and Pass Transistor logic based XOR and AND gates in the input side to enhance performance parameters and to reduce transistor count. Two new modified CLA adder have been proposed and their performance parameters have been compared with conventional circuit using Cadence simulation tools. The proposed designs showed significant performance improvement while keeping the advantages of static CMOS Logic.**

Keywords—carry look ahead, pass transistor, gate diffusion input, adder.

I. INTRODUCTION

With rapid progress in the field of science and technology, systems have become small and portable which requires low power circuitry within a very small area [1-30]. Therefore, designs having less transistor count and less surface area with suitable performance levels have become prominent [31]. However, according to conventional theory, delay and area are two parameters in chip design which are inversely related to each other [32]. Hence, it remains a challenging task for designers to enhance circuit performance, meanwhile keeping its surface area low.

Adders play a vital role in arithmetic circuits of modern microprocessors for which it is necessary to have optimized performance of adders [33]. Many Adder topology have been invented by various researchers: each adder having its own pros and cons [34], [35]. For wide adders, design of Parallel Prefix Adders (PPA) have emerged in the recent years in order to optimize circuit parameters and performance aspects [36]. Carry Look Ahead (CLA) Adder in static CMOS (S-CMOS) logic is one of the most widely used PPA due to its simplicity and performance levels [37]. In static CMOS design, pull-up network consists of p-channel CMOS (pMOS) and pull-down network consists of n-channel CMOS (nMOS) in order to produce full swing output [38]. Since, output gets directly connected either to power supply or ground in S-CMOS technique, driving capacity of such circuits are quite satisfactory. Moreover, S-CMOS is robust against transistor scaling and sizing for which this design technique is quite versatile. However, high transistor count in S-CMOS based design result in high area consumption on chip and high input impedance corresponds to increased delay [39], [40]. The XOR and AND gates in the input side of S-CMOS based CLA accounts for most of the transistor, input impedance and power. Thus, for further performance enhancement, it is necessary to redesign input side circuitry of S-CMOS based CLA while keeping the its advantages.

Lowering supply voltage with decreased clock speed might appear as a solution for less power consumption [41]. However, these increase delay significantly for which lowering supply voltage and clock can be applied to a certain extend. Use of fast gates might appear as another potential solution for delay improvement. But, due to limitations in physical components for circuit construction, logic gates can be only made faster to a certain extend [42], [43]. Hence, lowering voltage, clock speed decrement or applying faster gates is not a potential solution for performance enhancement. In addition, transistor count is not reduced if the techniques mentioned above are used. Thus, actual solution to enhance performance lies in applying new design techniques to existing ones in a proficient manner.

This paper presents two new designs to enhance performance parameters and reduce transistor count of S-CMOS based 4-bit CLA. Gate Diffusion Input (GDI) and Pass Transistor Logic (PTL) are used for designing the proposed CLA adders. Since 4-bit adders are used in many cases to design wide adders, performance of 4-bit adder plays a vital role in overall performance [44-46]. In order to analyze performance, simulations of the proposed designs and existing design have been carried out using Cadence tools in 90 nm

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technology. The proposed designs displayed significant improvement in terms of power and delay which account for low power delay product (PDP). Moreover, transistor count of the proposed designs is notably low which is responsible for making the circuit areas smaller.

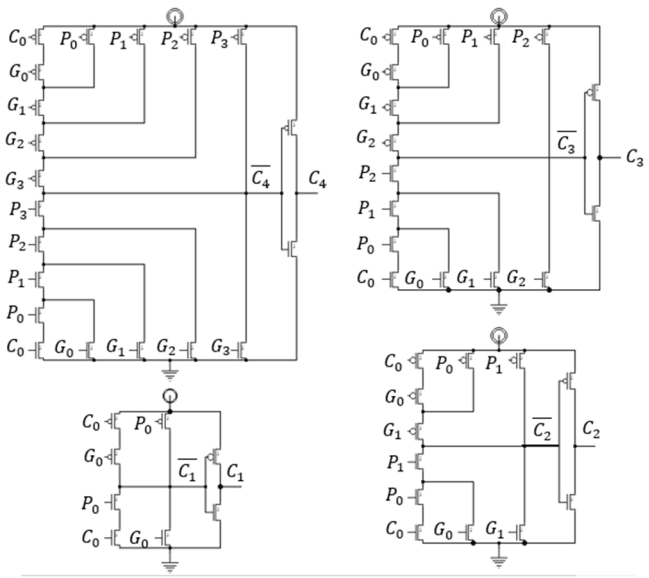


Fig. 1. Carry Generations Circuits of S-CMOS based CLA adder.

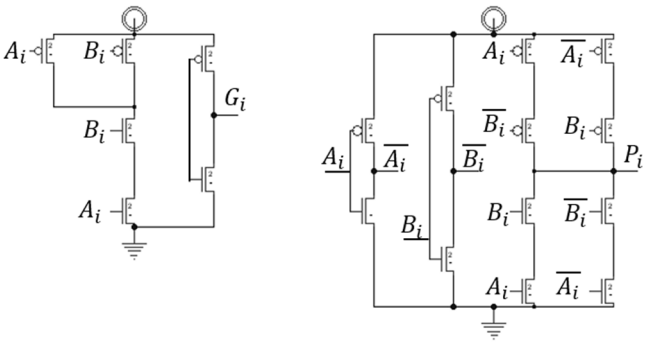


Fig. 2. S-CMOS based AND gate and XOR gate design.

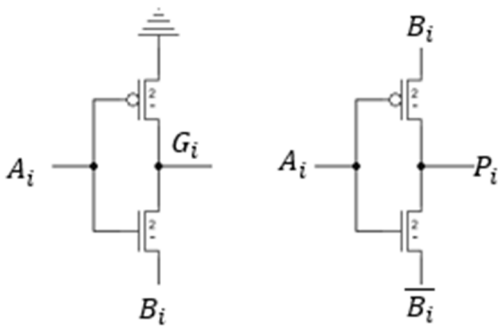


Fig. 3. GDI based AND and XOR gates for generating Gi and Pi terms in proposed design 1.

II. Existing S-CMOS Based CLA Design

According to [47], circuit diagrams for carry-out bits C_4, C_3, C_2 and C_1 are presented in Fig. 1. Here, the term G_i represents $A_i B_i$ (A_i AND B_i) and the term P_i represents $A_i \oplus B_i$ (A_i XOR B_i), where the term i = 0, 1, 2, 3 or 4. Thus, Formula for carry generation is:

$$C_{i+1} = G_i + P_i C_i$$

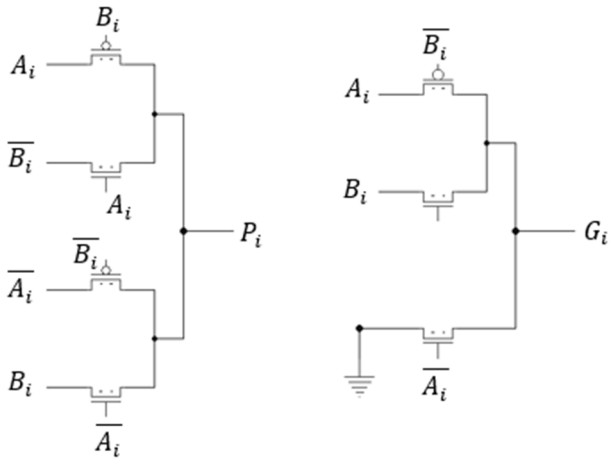


Fig. 4. AND gate and XOR gate which are used for generating G_i and P_i terms in proposed design 2 [49].

XOR gate and AND gate design in S-CMOS logic, which are used in conventional S-CMOS based CLA in [47] are represented by Fig. 2. The XOR gate in S-CMOS design requires 12 transistors and AND gate requires 6.

Formula for sum generation is as follows:

$$S_i = P_i \oplus C_i$$

III. Proposed Designs

Design descriptions for three proposed CLAs are provided in the following sub-sections.

A. Design 1: GDI and S-CMOS based 4-bit CLA

Gate Diffusion Technique (GDI) allows logic design with very low transistor count [46]. Design of basic logic gates using GDI technique is reported in [47], [48]. Design of AND gate and XOR gate in GDI technique is represented by Fig. 3.

GDI gates do not provide full swing output for which voltage degradation is a major problem with GDI based design. Hence, GDI based XOR gates and AND gates are used in input side for generating the P_i and G_i terms. These P_i and G_i terms are fed into S-CMOS based carry generation circuits presented in Fig. 1 for proving full swing output. For generating sum, the G_i terms are fed into S-CMOS based XOR gates presented in Fig. 2 which also provides full swing logic. Thus, final outputs of this GDI and S-CMOS based 4-bit CLA design provides full swing logic and the problem of voltage degradation is removed.

B. Design 2: PTL and S-CMOS based 4-bit CLA

In this design, for generating G_i and P_i terms, PTL AND gates and XOR gates used for designing in Dual Pass

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Transistor Logic adder (DPL) are used [49]. The AND and XOR logic gates used for DPL adder are represented by Fig. 4. These gates have low transistor count and low input impedance. However, driving capability of such sort of PTL gates are quite low. Since, S-CMOS based carry generation circuits and XOR gates presented in Fig. 1 and Fig. 2 respectively are used in output side. Driving capability of this proposed 4-bit CLA adder is quite good due to having S-CMOS in the output side.

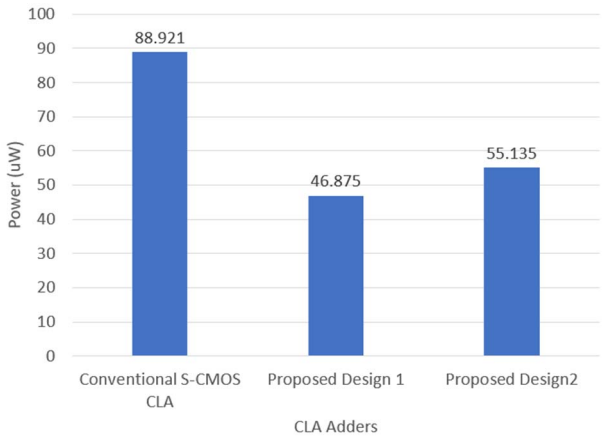


Fig. 5. Power consumption of conventional design and proposed designs.

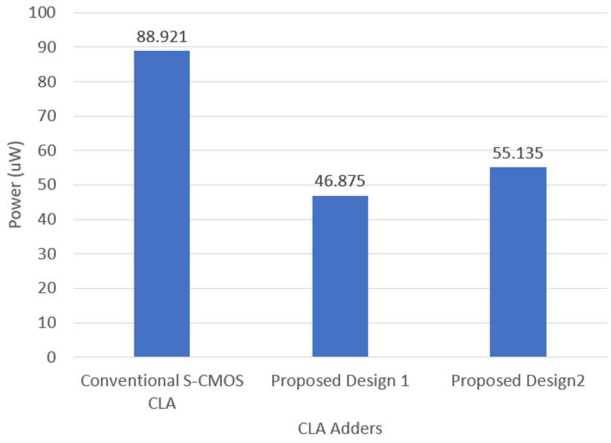


Fig. 6. Delay for conventional design and proposed designs.

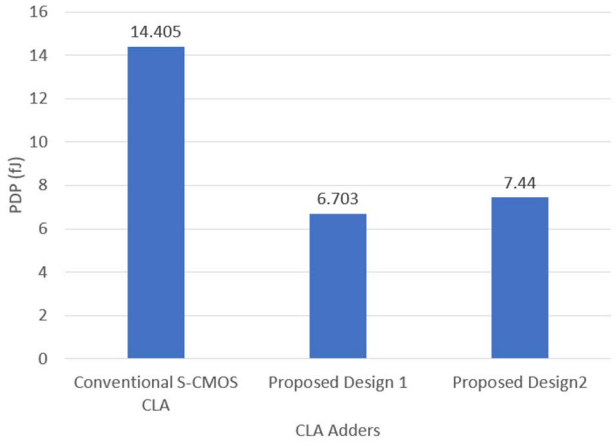


Fig. 7. PDP for conventional design and proposed designs.

IV. Simulation Result

In order to compare the proposed designs with conventional S-CMOS based design, simulation was carried out using Cadence Virtuoso Tools. With the aim to maintain common environment for all the designs (proposed and existing), 90 nm technology was chosen and the supply voltage for all circuits was 1.2 V.

The simulated data on power, delay and power delay product (PDP) of the two proposed designs and existing S-CMOS based CLA design is represented in Table I. The data listed in Table I are also presented in graphical manner so that it gets easier to get an overall view of Table I. Performance parameters improvements of the proposed designs compared to the conventional design are listed in Table II. Transistor count of the CLA adders (proposed and conventional) are presented in Table III.

TABLE I. Simulation Result of Proposed and Conventional Designs

CLA Adder Design	Ref no.	Power (µW)	Delay (ns)	PDP (fJ)
Conventional S-CMOS	[7]	88.921	0.162	14.405
Proposed Design 1	Present	46.875	0.143	6.703
Proposed Design 2	Present	55.135	0.135	7.44

TABLE II. Improvements in Performance Parameters of The Proposed Designs

CLA Adder Design	Improvement in Power	Improvement in Delay	Improvement in PDP
Proposed Design 1	47.28 %	11.728 %	53.468 %
Proposed Design 2	37.996 %	16.67 %	48.351 %

TABLE III. Transistor Count of Conventional S-CMOS Based CLA and Proposed CLA designs

CLA Adder Design	Ref no.	Transistor Count
Conventional S-CMOS	[7]	176
Proposed Design 1	Present	106
Proposed Design 2	Present	142

V. Comparative Analysis and Discussion

From the simulation result listed in Table I, it can be easily observed that the proposed designs made noteworthy improvement in terms of power, delay and PDP. In is observed that proposed design 1 made more improvement in terms of power than proposed design 2. But in terms of speed (delay), improvement of proposed design 2 is higher than proposed design 1. Hence, design 1 is more suitable for low power applications whereas design 2 is suitable for systems that require fast response.

In terms of transistor count, both of the proposed designs have significantly less transistors compared to conventional S-CMOS based CLA adder in [37]. Low transistor count accounts for low surface area on chip which allows circuits be become more compact. Thus, the proposed designs are much more suitable in chip designing than the existing design. However, design 1 made more transistor count reduction than design 2.

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According to [50], rather than sticking to only one logic technique, it is better to utilize the advantages of several logic techniques in one circuit for optimized performance levels. Although S-CMOS based circuits have ample driving capacity, but high input impedance makes the circuits slow. In order to reduce input impedance, the proposed designs used AND and XOR gates having very low transistor count in the input side. This modification resulted in making the proposed designs faster. However, no modification has been made in the output side circuits which generate carry terms and sum terms. Reason for this is to preserve the driving capability using S-CMOS logic. Hence, the modified CLA adder designs proposed in this paper reduced the drawbacks of conventional design while preserving its advantages.

VI. Conclusion

In this research, two designs of CLA adder have been proposed with an aim to reduce transistor count and escalate performance parameters. To verify the performance aspects of the proposed work, simulations were performed in Cadence environment. Modifications done to conventional S-CMOS based design of CLA adder showed noteworthy performance development. The proposed designs lessened the problems in conventional design while keeping its advantages the same. Hence, utilizing the proposed designs in arithmetic circuits of modern systems will definitely make the system better [51-53].

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